Mock central exam: VWO wiskunde (B / techniek style)

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An original practice paper written in the style of the Dutch central exam (*centraal examen*, CE) for VWO Wiskunde B, with the additions of wiskunde techniek (parametric curves, a differential equation, an algorithm, probability, a geometric proof). It is not an official exam.

Time: 3 hours (180 minutes) **Total:** 75 points, 19 questions in 8 parts **Allowed:** graphing calculator (*grafische rekenmachine*) in exam mode, ruler, compass, protractor. No formula sheet other than the one on the last page.

Instructions (as on the real exam):

- Write down your working (*berekeningen, toelichting, redenering*). An answer without working usually earns no points.
- *Bereken exact* / *exact*: no decimal approximations; the calculator may only be used to check.
- *Toon aan* / *bewijs*: the result is given; you must show every step that leads to it.
- *Benader* / *bereken* (without "exact"): a calculator result is allowed, but write down what you entered (the function, the equation and the window or interval).
- Round only at the end, to the precision asked.

Do this paper in one sitting under exam conditions. Mark it with the solutions and record every lost point in your error log.

Part 1: A function with an exponential

The function f is given by $f(x) = x\,e^{-\frac{1}{2}x}$.

1 (4p) The graph of f has a maximum. *Bereken exact* the coordinates of the top (*top*) of the graph.

2 (4p) V is the region bounded by the graph of f, the x-axis and the line $x = 2$. *Bereken exact* the area of V.

3 (5p) The graph of f has exactly one point of inflection (*buigpunt*). *Toon aan* that the tangent line at this point of inflection meets the x-axis at $(8, 0)$.

Part 2: A trigonometric function

On the domain $[0, 2\pi]$ the function g is given by $g(x) = \sin 2x + 2\cos x$.

4 (5p) *Toon aan* that the maximum value of g is $\frac{3}{2}\sqrt{3}$.

5 (3p) *Bereken exact* the zeros of g (*nulpunten*).

6 (4p) The function $h(x) = 2\sin x$ on $[0, \pi]$ together with the x-axis bounds a region W. W is rotated about the x-axis. *Bereken exact* the volume of the solid of revolution (*omwentelingslichaam*).

Part 3: A circle and two tangent lines

The circle c is given by $x^2 + y^2 - 6x - 4y + 8 = 0$. The lines ℓ_k are given by $y = 2x + k$.

7 (4p) *Bereken exact* the values of k for which ℓ_k is tangent to c.

The points $P(1,1)$ and $Q(4,4)$ lie on c. The tangent lines to c at P and at Q intersect in a point T.

8 (4p) Calculate (*bereken*) the angle $\angle PTQ$ in degrees, rounded to a whole number.

Part 4: A Lissajous curve

A point moves along a curve K given by

$$x(t) = \cos t, \qquad y(t) = \sin 2t, \qquad 0 \le t \le 2\pi.$$

9 (4p) *Bereken exact* the coordinates of all points of K where the tangent line is horizontal.

10 (4p) The curve passes through the origin twice and crosses itself there. Calculate the angle between the two tangent directions at the origin, in degrees, rounded to a whole number.

11 (3p) *Toon aan* that every point of K satisfies $y^2 = 4x^2(1 - x^2)$.

Part 5: Filling a tank

A tank is filled with water at a constant 3 litres per minute while it leaks at a rate of $0.1V$ litres per minute, where V is the volume (litres) at time t (minutes). At $t = 0$ the tank is empty. So

$$\frac{dV}{dt} = 3 - 0.1V, \qquad V(0) = 0.$$

12 (3p) *Toon aan* that $V(t) = 30\left(1 - e^{-0.1t}\right)$ is the solution of this differential equation with this initial condition.

13 (4p) Use Euler's method with step size $\Delta t = 2$ to approximate $V(4)$. Then calculate the difference between your approximation and the exact value $V(4)$, rounded to two decimals.

Part 6: Bolts

A machine produces bolts. The diameter D (mm) is normally distributed with mean $\mu = 10.00$ and standard deviation $\sigma = 0.05$. A bolt is rejected (*afgekeurd*) if its diameter is less than 9.90 or more than 10.10 mm.

14 (3p) Calculate the probability that a randomly chosen bolt is rejected. Round to three decimals.

15 (4p) A box contains 50 bolts. Assume the bolts are independent. Calculate the probability that at least 3 bolts in the box would be rejected. Round to three decimals.

Part 7: A dice game

At a fair you pay €1 to throw two fair dice once. If you throw a double you receive €4; if the sum of the eyes is 7 you receive €1; otherwise you receive nothing. Let G be your net gain (*winst*) in euros for one game, taking the €1 stake into account.

16 (4p) Calculate the expected value $E(G)$ and the standard deviation $\sigma(G)$. Round $\sigma(G)$ to two decimals.

Part 8: Families and proofs

For $p > 0$ the functions f_p are given by $f_p(x) = \ln(x^2 + p)$.

17 (5p) Each graph of f_p has two points of inflection. *Toon aan* that, for every $p > 0$, these points of inflection lie on the graph of $y = \ln(2x^2)$.

For every real a the function k_a is given by $k_a(x) = x^3 - ax$.

18 (4p) *Toon aan* that the tangent lines to the graphs of k_a at $x = 1$ all pass through one point, and give that point.

$ABCD$ is a cyclic quadrilateral (*koordenvierhoek*). The lines AB and DC intersect in a point S outside the circle, with B between A and S and C between D and S.

19 (4p) *Bewijs* that triangle SBC is similar to triangle SDA, and deduce that $SB \cdot SA = SC \cdot SD$.

Formula sheet

Product rule	$(fg)' = f'g + fg'$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
Chain rule	$(f(g(x)))' = f'(g(x))\, g'(x)$
Volume of revolution about the x-axis	$\pi \int_a^b f(x)^2\, dx$
Distance from (p, q) to $ax + by = c$	$\frac{\lvert ap + bq - c\rvert}{\sqrt{a^2 + b^2}}$
Angle between vectors	$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{\lvert\vec{u}\rvert\, \lvert\vec{v}\rvert}$
Trigonometry	$\sin 2x = 2 \sin x \cos x$, $\cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1$
Expected value, variance	$E(X) = \sum x\, P(X = x)$, $\mathrm{Var}(X) = E(X^2) - E(X)^2$
Euler's method	$V_{n+1} = V_n + \Delta t \cdot V'(t_n)$